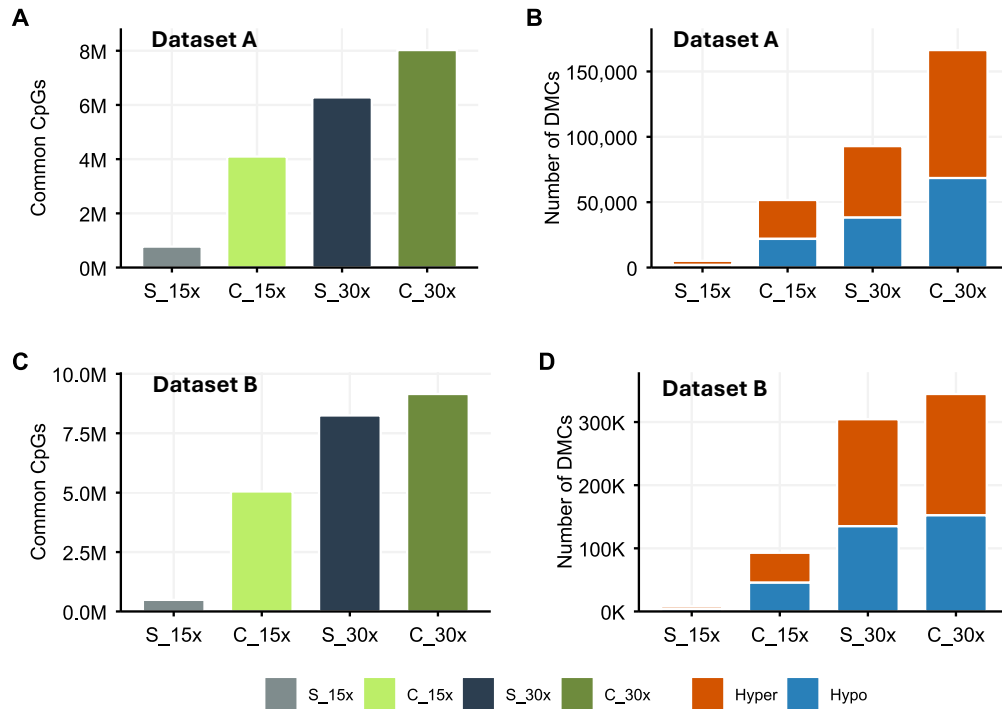
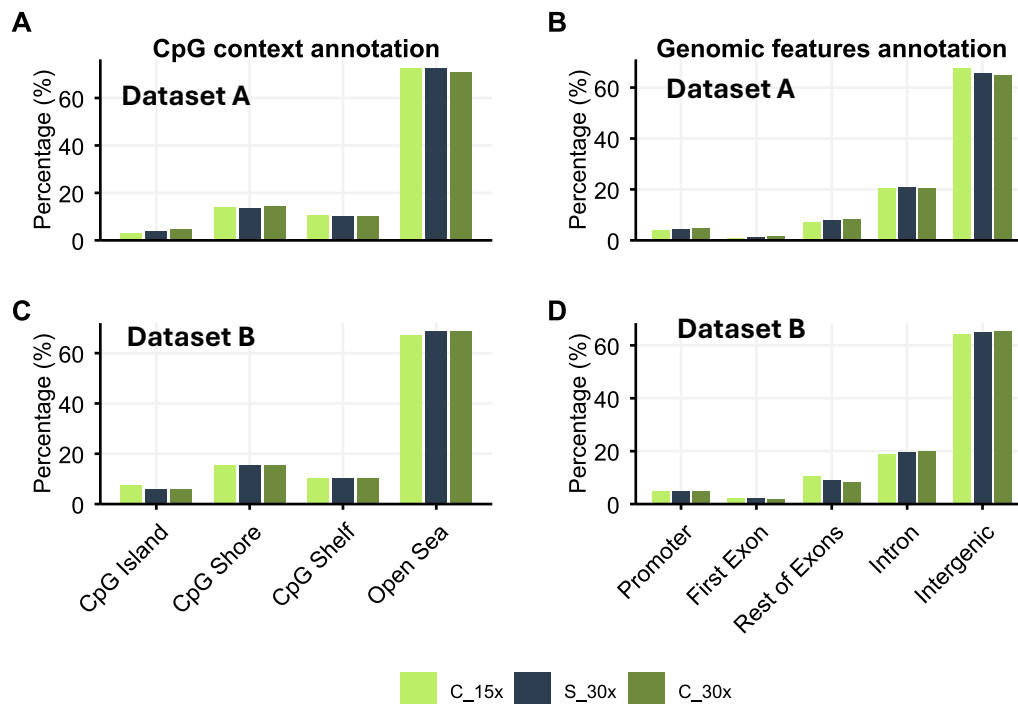


## Supplementary Material

### Supplementary Figures



*Supplementary Figure 1. Common CpG sites between samples with the different methodologies and a coverage threshold of 5×. Standard methylKit (S) and CovGain (C). A) Common CpG sites between AT and EM samples of the dataset A (n = 7) at different depths (15× and 30×) with S and C. B) Hyper- and Hypo-methylated CpG sites between AT and EM samples of the dataset A (n = 7) at different depths (15× and 30×) with S and C. C) Common CpG sites between AT and EM samples of the dataset B (n = 24, 25% missing data) at different depths (15× and 30×) with S and C. D) Hyper- and Hypo-methylated CpG sites between AT and EM samples of the dataset B (n = 24, 25% missing data) at different depths (15× and 30×) with S and C.*



*Supplementary Figure 2. Comparative analysis of CpG context and Genomic Features annotation across the different methodologies and datasets. (A-D) Distribution of CpG sites across genomic categories for Dataset A (A and B) and Dataset B (C and D). Bars represent the percentage of total CpG sites identified for each genomic feature and by each method: C\_15x, S\_30x, C\_30x.*

## Supplementary tables

**Supp. Table 1:** Sequencing statistics of the Dataset A at 15x and 30x sequencing depth.

**Supp. Table 2:** Sequencing statistics of the Dataset B at 30x sequencing depth.

**Supp. Table 3:** CpG-context DNAm information summary for the Dataset A

**Supp. Table 4:** ANOVA results from the comparisons of the per-sample DNAm information of the Dataset A.

**Supp. Table 5:** Pairwise Tukey HSD test from the comparisons of the per-sample DNAm information of the Dataset A.

**Supp. Table 6:** CpGs and DMCs identified across methodologies and sequencing depths for both datasets.

**Supp. Table 7:** CpG context and genomic feature distribution of the CpGs analysed across methodologies and sequencing depths for Dataset A.

**Supp. Table 8:** CpG context and genomic feature distribution of the CpGs analysed across methodologies and sequencing depths for Dataset B.